

Check tests according to table and review results below. Results do not have to be available to start patient on ART same day. Record correct contact details in case of abnormal results to recall patient.

At diagnosis	Starting/changing ART	After 1 month on ART	After 3 months on ART	After 10 months on ART	6 monthly	Yearly	Also
• Urine: dipstick and pregnancy test¹ • Sputum: TB NAAT • Blood: creatinine, Hb, HBsAg, CD4 (and CrAg² if CD4 < 100) • Cervical screening	• Changing to TDF: creatinine • Changing from TDF: HBsAg • Starting AZT: Hb	AZT: FBC + diff	• Viral load • TDF: creatinine • AZT: FBC + diff • ATVr or LPVr: total cholesterol, triglycerides	• Viral load • CD4 • TDF: creatinine	If previous CD4 < 200 or not on ART: CD4	• Viral load • TDF: creatinine • Sputum: TB NAAT	• Check viral load more often : - Pregnant: at 1st antenatal visit and delivery - Breastfeeding: 6 monthly - RR-TB: 6 monthly • Cervical screen 3 yearly
ABC - abacavir ATVr - atazanvir/ritonavir AZT – zidovudine CrAg - cryptococcal antigen Diff - differential white cell count EFV - efavirenz FBC – full blood count FTC - emtricitabine Hb - haemoglobin HBsAg – hepatitis B surface antigen LPVr – lopinavir/ritonavir RR-TB – rifampicin-resistant TB TDF – tenofovir TEE - TDF + FTC + EFV TLD - TDF + 3TC + DTG 3TC - lamivudine							
Urine dipstick	• If proteinuria, check creatinine (eGFR) if not already done. Interpret result below. If pregnant, recheck urine dipstick, do BP and manage further ➔ 161. • If glucose in urine: check random fingerprick glucose ➔ 17.						TB tests changing from 'Xpert Ultra' to 'TB NAAT' (NAAT = nucleic acid amplification test and includes Xpert as well as newer TB tests).
Urine pregnancy test	• If pregnancy test positive, give antenatal care ➔ 159 and if not on ART, start same day. • If pregnancy test negative, advise to use reliable contraception (IUD, subdermal implant or sterilisation, <i>plus</i> condoms).						
TB sputum test	Interpret sputum TB NAAT results ➔ 92. Repeat TB sputum test yearly, at same time as yearly viral load done.						
CD4	If CD4 < 100, check CrAg result. If CD4 < 200 or WHO stage 2, 3 or 4 disease at HIV diagnosis, start co-trimoxazole prophylaxis therapy (CPT) ➔ 113 and do a rapid urine LAM test for TB if TB symptoms ➔ 92.						
CrAg ²	If cryptococcal antigen (CrAg) positive, refer for lumbar puncture (LP). If symptomatic (headache, confusion) or pregnant, refer urgently.						
Hb (FBC + differential count)	• If Hb < 12 (woman) or < 13 (man), anaemia likely, do FBC and differential count if not already done ➔ 27. If difficulty breathing, chest pain or dizziness, refer same day. • If Hb ≤ 8 or neutrophils ≤ 1.0: avoid zidovudine. If already on zidovudine, switch to TDF or ABC. If on AZT because of kidney problem and ABC hypersensitivity, discuss with HIV hotline ➔ 178.						
Hepatitis B (HBsAg)	• If HBsAg positive, TDF should form part of ART regimen ➔ 120. • If HBsAg negative, check immune response and give 3 doses of hepatitis B vaccine if needed ➔ 120.						
Cervical screen	Interpret result ➔ 55. Repeat 3 yearly if normal.						
Creatinine (eGFR)	• If eGFR < 30, refer/discuss with HIV hotline ➔ 178. • If eGFR ≤ 50 (or creatinine > 85 in pregnant patient): recall patient. Switch ART according to HbsAg result: - If HBsAg negative: stop TDF, use ABC instead. If on TLD or TEE: switch to ABC + 3TC + DTG. If previous hypersensitivity to ABC, use AZT instead of ABC. - If HBsAg positive: discuss management with experienced ART clinician or HIV hotline ➔ 178. • Check if other medication doses need adjusting: eGFR can be used as acceptable estimate of creatinine clearance (CrCl). Check for proteinuria and repeat eGFR after 1 month. If repeat eGFR ≤ 50, refer to doctor to check BP, glucose, urine dipstick, send urine for protein/creatinine ratio and arrange kidney ultrasound.						
ALT	• If ALT ≥ 200 or jaundice, stop medications and discuss/refer same day. • If ALT < 200: - If no symptoms, continue medications and monitor for symptoms. Also repeat ALT weekly until < 120. - If symptoms (nausea/vomiting/abdominal pain): • If ALT 120-199 (or total bilirubin > 40, if done): stop all medications and discuss/refer same day. • If ALT 50-120: doctor to assess for causes (check HBsAg; consider alcohol or drug-induced liver injury), consider interrupting/delaying ART. Repeat ALT within 1 week. If unsure, discuss with specialist.						
Total cholesterol, triglycerides	• If triglycerides ≥ 10, discuss/refer same day. • If total cholesterol > 6 or triglycerides > 5, and on LPVr or ATVr, switch to DTG ➔ 117. If unable to switch to DTG, switch to ATVr and change simvastatin to atorvastatin 10mg at night. Repeat fasting total cholesterol and triglycerides after 3 months: if still raised, discuss/refer.						
Viral load (VL)	• If VL < 50, continue routine VL monitoring (see table above). If not yet on TLD, switch ART ➔ 117. Check if eligible to collect medications from a repeat prescription collection point ➔ 113. • If VL ≥ 50, manage unsuppressed viral load ➔ 119.						
Advise and treat the patient with HIV ➔ 113.							

¹Only do pregnancy test if woman of child bearing potential has missed period and is not on contraception. ²CrAg - cryptococcal antigen. Laboratory will usually automatically do this if CD4 < 100.